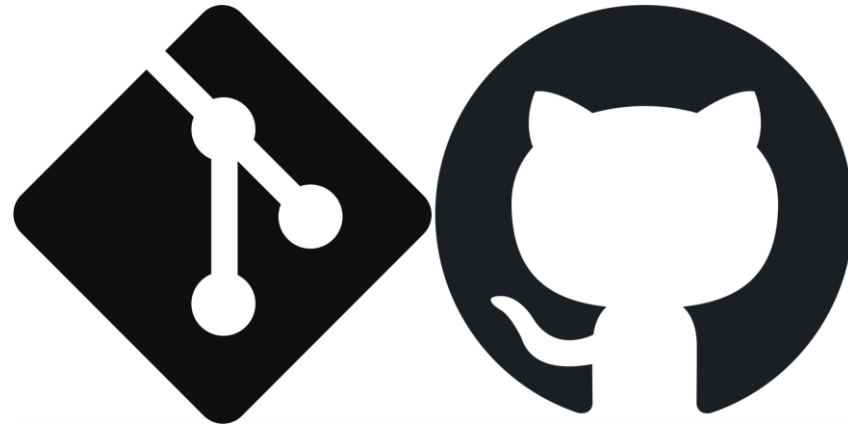




Introduction to Git & GitHub

Jennifer Graham

SAMS Session 3: Using GitHub to collaborate

12th May 2026

Topics to cover today

- Publishing code
 - Releases & DOIs
 - README & Wiki documentation
- GitHub tools
 - Issues
 - Projects
- Collaborative code development
 - Permissions, forking vs branching?
 - Pull requests?
 - GitHub Organisations

Recap from Sessions 1 & 2 ...

So, what are Git & GitHub?

NB. These are two different things

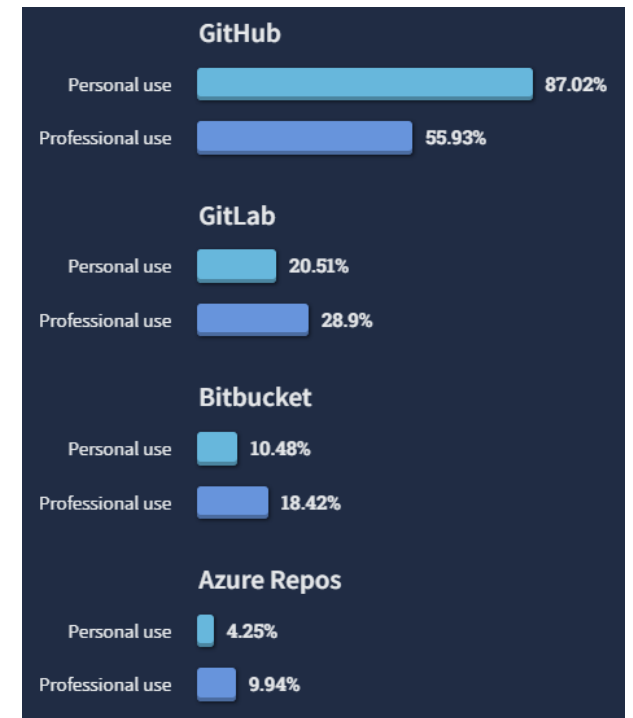
- Git = version control software.
- GitHub = web-based repository hosting service*.

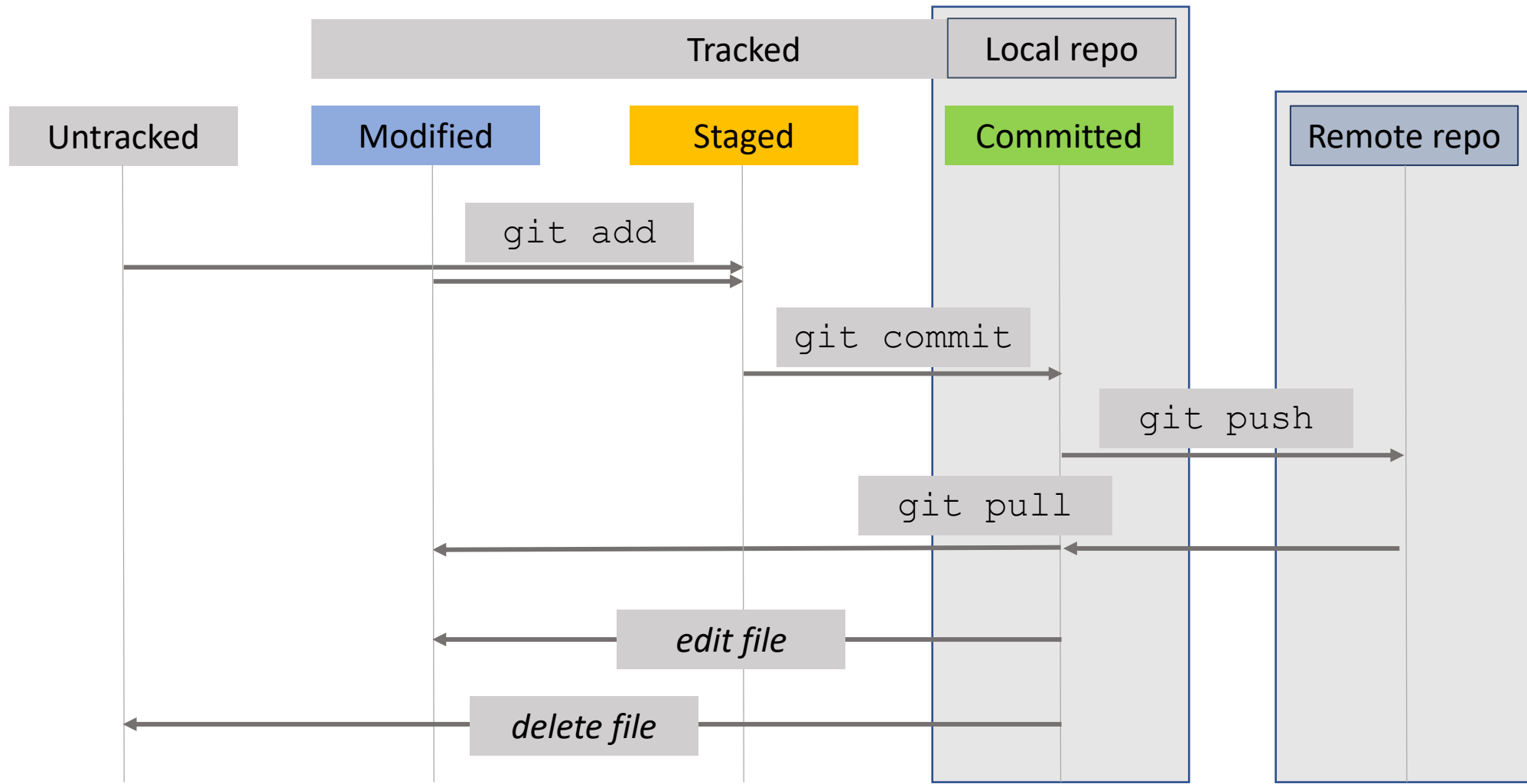
i.e. Git is the software behind the GitHub web service.

You can use Git without GitHub. However, GitHub provides useful tools (especially for sharing your code).



* Other providers exist, but GitHub is the most popular, e.g.:





<code>git init</code>	:: turn current directory into git repository.
<code>git add <file></code>	:: stage the file for commit i.e., track changes.
<code>git commit</code>	:: confirm changes with message/explanation.
<code>git push</code>	:: publish/back-up changes on GitHub (remote server).
<code>git pull</code>	:: update your local repository with remote changes, from GitHub (remote server).
<code>git status</code>	:: show which files have been tracked or modified.
<code>git diff <file></code>	:: show difference between current file and last commit.
<code>git log <file></code>	:: show commit history [for file]

How to roll back changes?

```
git restore <path>
```

:: discard changes in working directory for chosen file(s).

```
git checkout <commit> [--] <path>
```

:: retrieve version of file(s) from chosen commit.

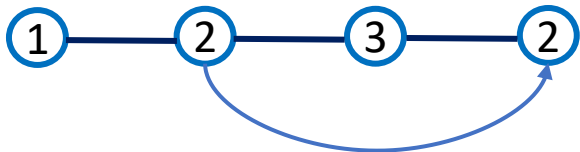
```
git revert <commit>
```

:: undo your chosen commit, keeping record of change.

```
git reset [--soft/hard] <commit>
```

:: discard all commits [and possibly changes!] since chosen version
- use with care!!!

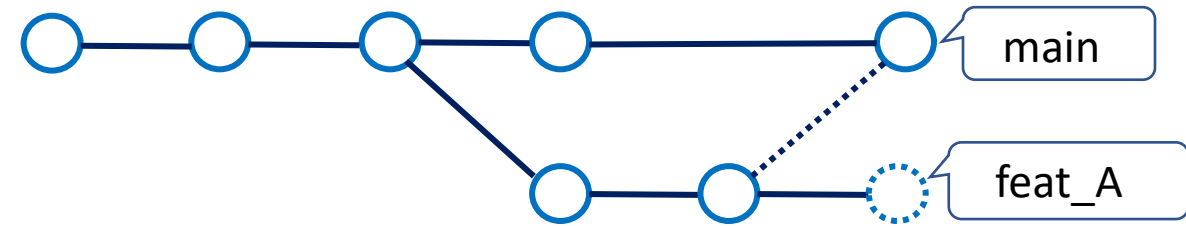
git revert commit3



git reset --soft/hard commit2



Branches: How to create and merge?



```
git branch    :: list existing branches; * = you are here.
```

```
git branch <branch-name>      :: create a new branch, <branch-name>.
```

```
git checkout <branch-name> :: "switch" onto branch, <branch-name>.
```

```
git switch [-c] <branch-name>
```

```
:: switch to your branch, <branch-name>,
```

```
-c option = create if new (shortcut for git branch and checkout).
```

```
git merge <branch-name>
```

```
:: apply new commits from <branch-name> (since split), onto present branch
```

```
git diff <branch1>..<branch2> [<path>]
```

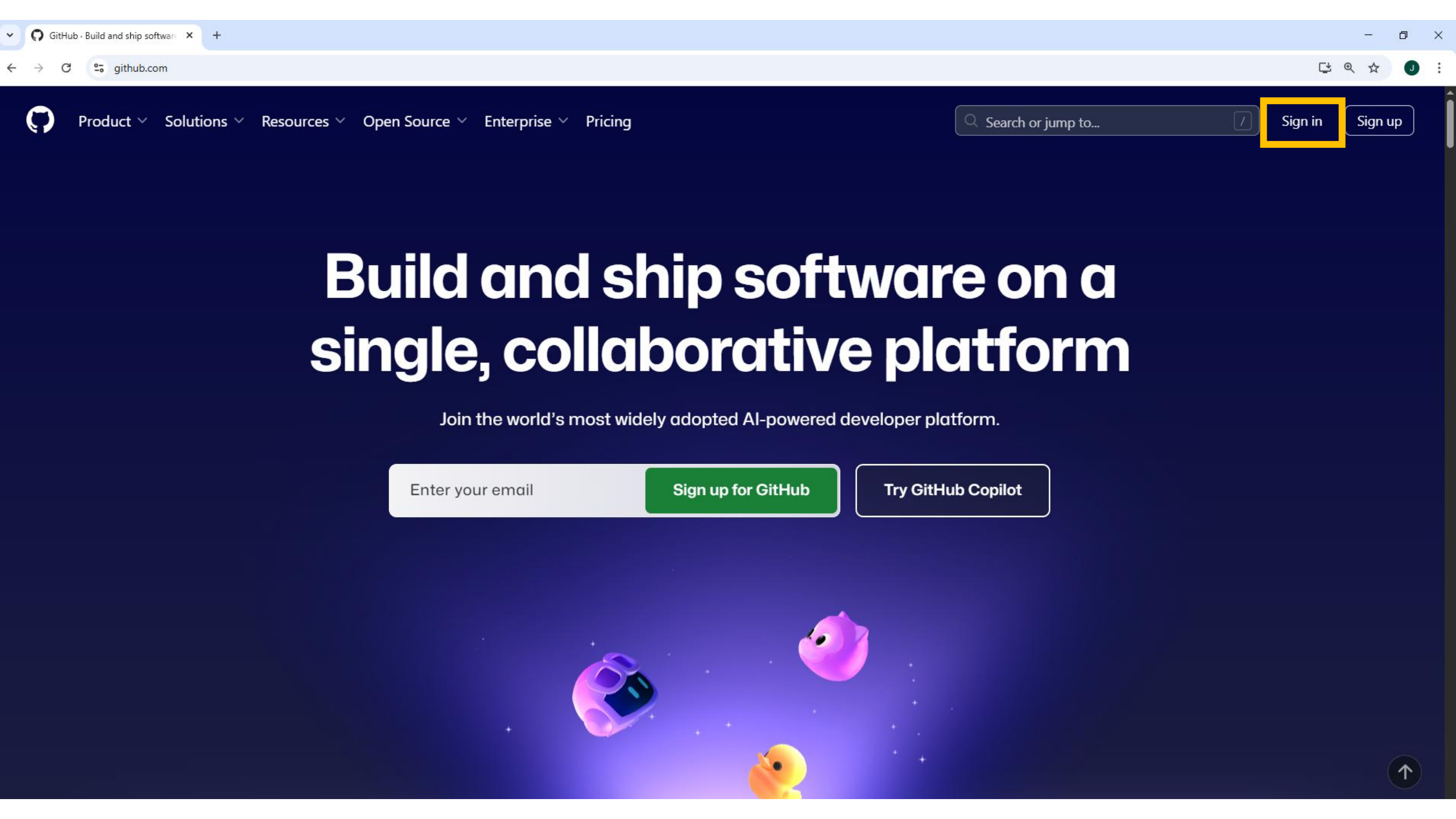
```
:: check difference between two branches, for optional file(s) <path>
```


Publishing code on GitHub

Releases & DOIs

README

Wiki



[Product](#) [Solutions](#) [Resources](#) [Open Source](#) [Enterprise](#) [Pricing](#)

Search or jump to...

Sign in

Sign up

Build and ship software on a single, collaborative platform

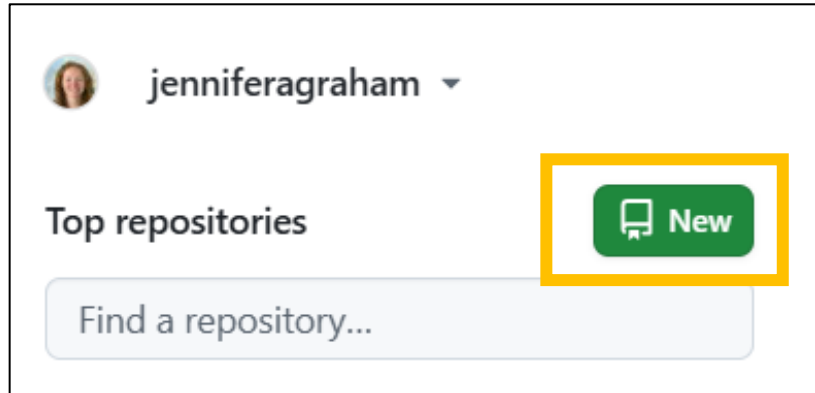
Join the world's most widely adopted AI-powered developer platform.

Enter your email

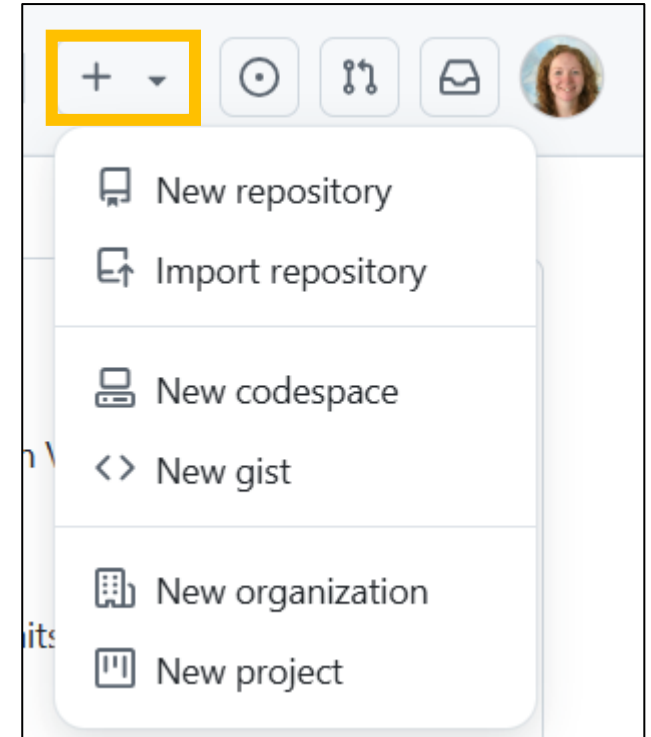
Sign up for GitHub

Try GitHub Copilot





Options to create your new repository.



Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner *

 jenniferagraham

Repository name *

hello_world

✓ hello_world is available.

Great repository names are short and memorable. How about [didactic-memory?](#)

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

Add license

Licenses explain how others can use your code. [About licenses](#)

 Public

✓  Public

Anyone on the internet can see this repository. You choose who can commit.

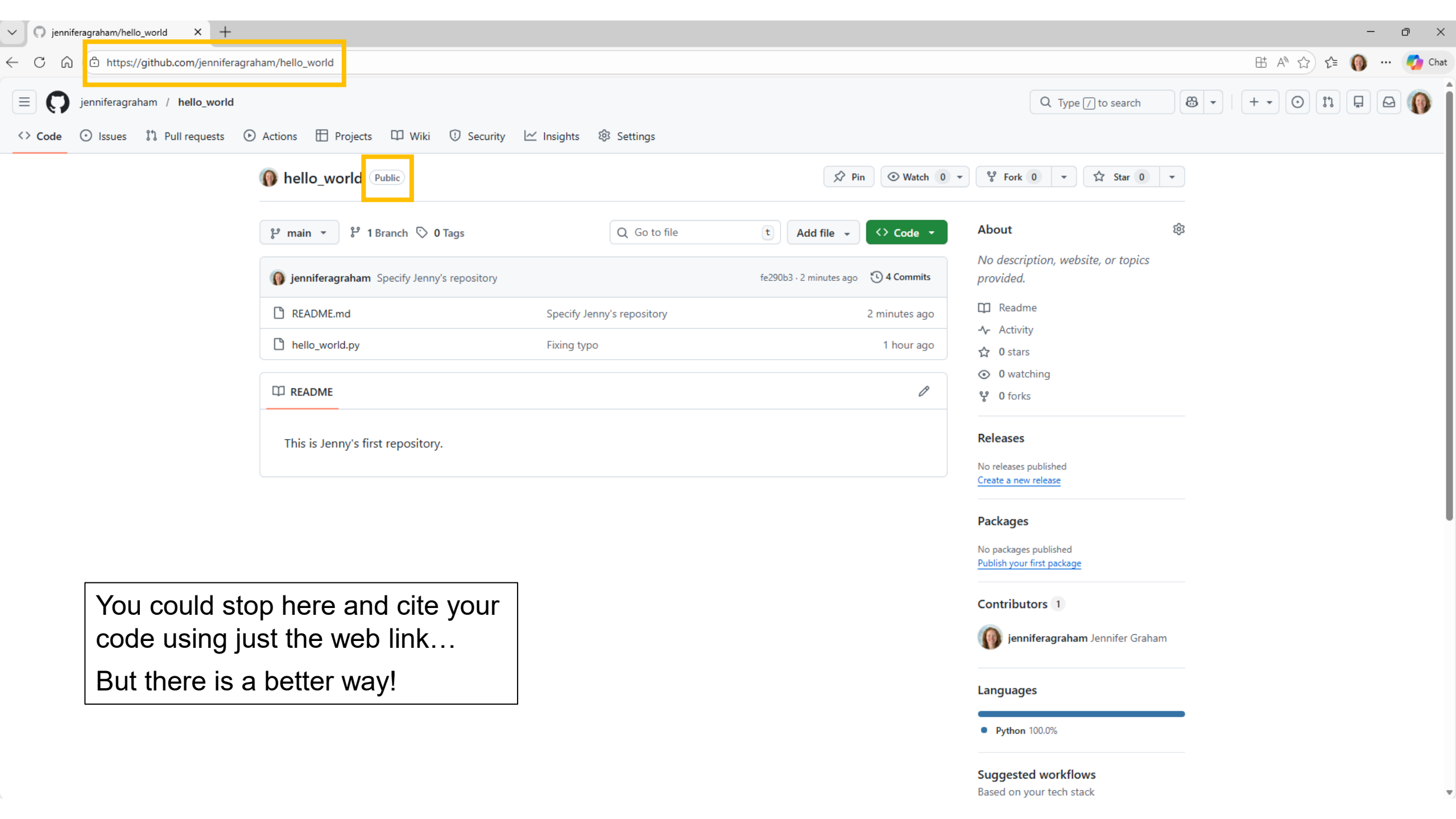
 Private

You choose who can see and commit to this repository.

No license

Create repository

NB. GitHub's default is public, but you may want to ensure control over visibility.



You could stop here and cite your code using just the web link...
But there is a better way!

Before you publish...

Adding Documentation:

README

Wiki

Adding documentation

There are many ways you can provide documentation on what your code does and how someone else can use it.

- **README**

- Most useful as it is automatically displayed on the front page of your repository.
- [Markdown](#) formatting can be used
- NB. Make sure to refer to any license (or restrictions) for use.

- **Wiki**

- Broader information on use of repository.
- Only available within public repositories or organisations.
- e.g. https://github.com/jenniferagraham/Git_Training/wiki/

- **Issues**

- Can flag development stages or known bugs/problems.

- **Projects**

- Organise ongoing or planned work (related issues) on repositories?

- NB. Some repository features are only available within organisations or paid accounts

- More info here: <https://github.com/pricing>

NB. Your commit messages are still an important way to document the code development.

README files

GitHub will automatically display your README on the front page of the repository.

- [Markdown](#) formatting can be used.
- Can be a simple text file, or any other text language?
- You can have new README files in each subdirectory.

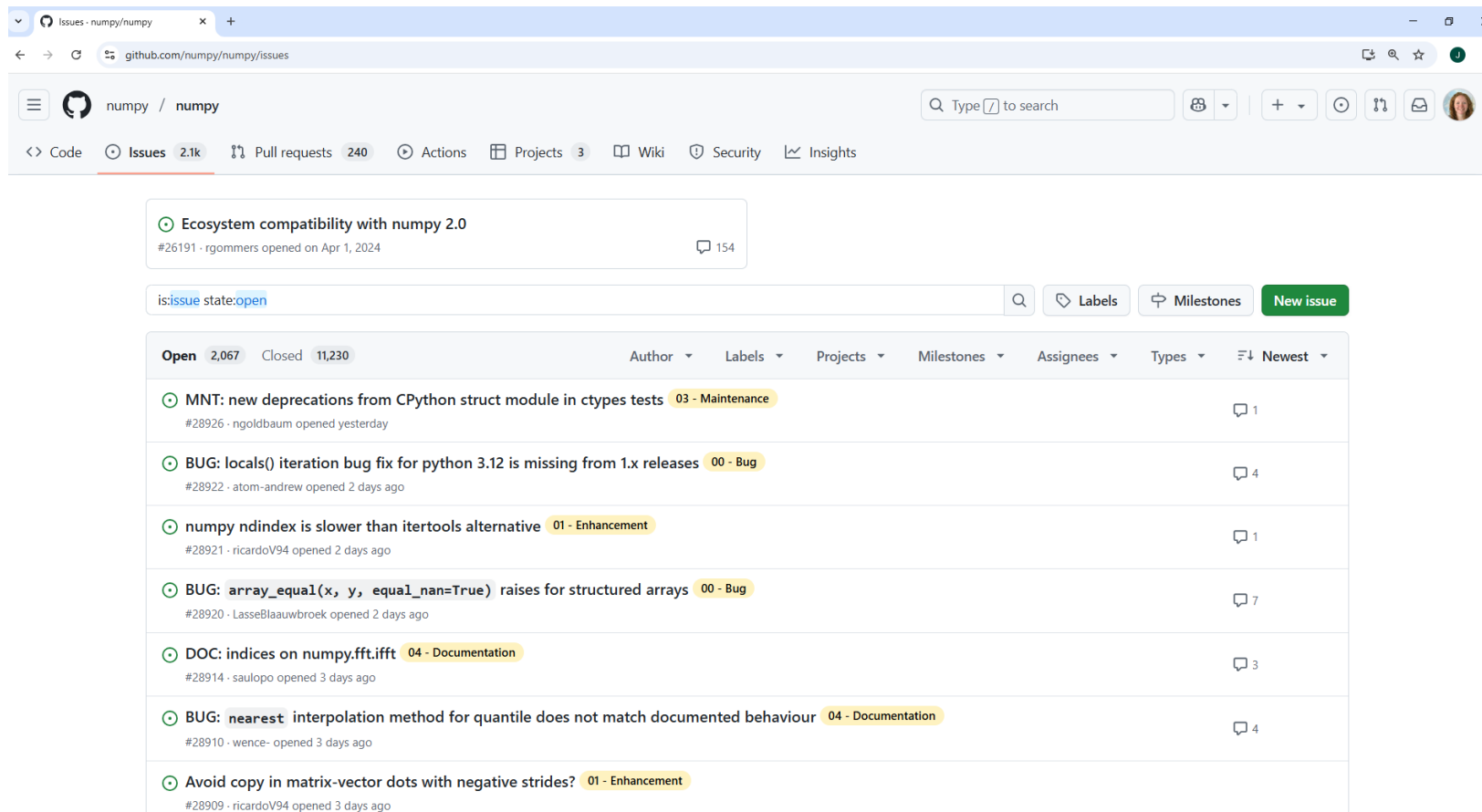
Wiki pages

Can add multiple pages of documentation.

- [Markdown](#) formatting will be used, easy to edit with previews online.
- NB. Wiki files are not tracked in the core repository.
 - If prefer to edit locally can clone as separate repository: <repository-name>.wiki
- For standard user accounts, wiki pages are only available for public repositories.
- They are available for use in private repositories, under organisation (or paid) accounts.

Issues

- A key feature for developers, useful to flag features, maintenance, or known problems/bugs.
- Issue numbers can be cited elsewhere e.g. in commit messages.

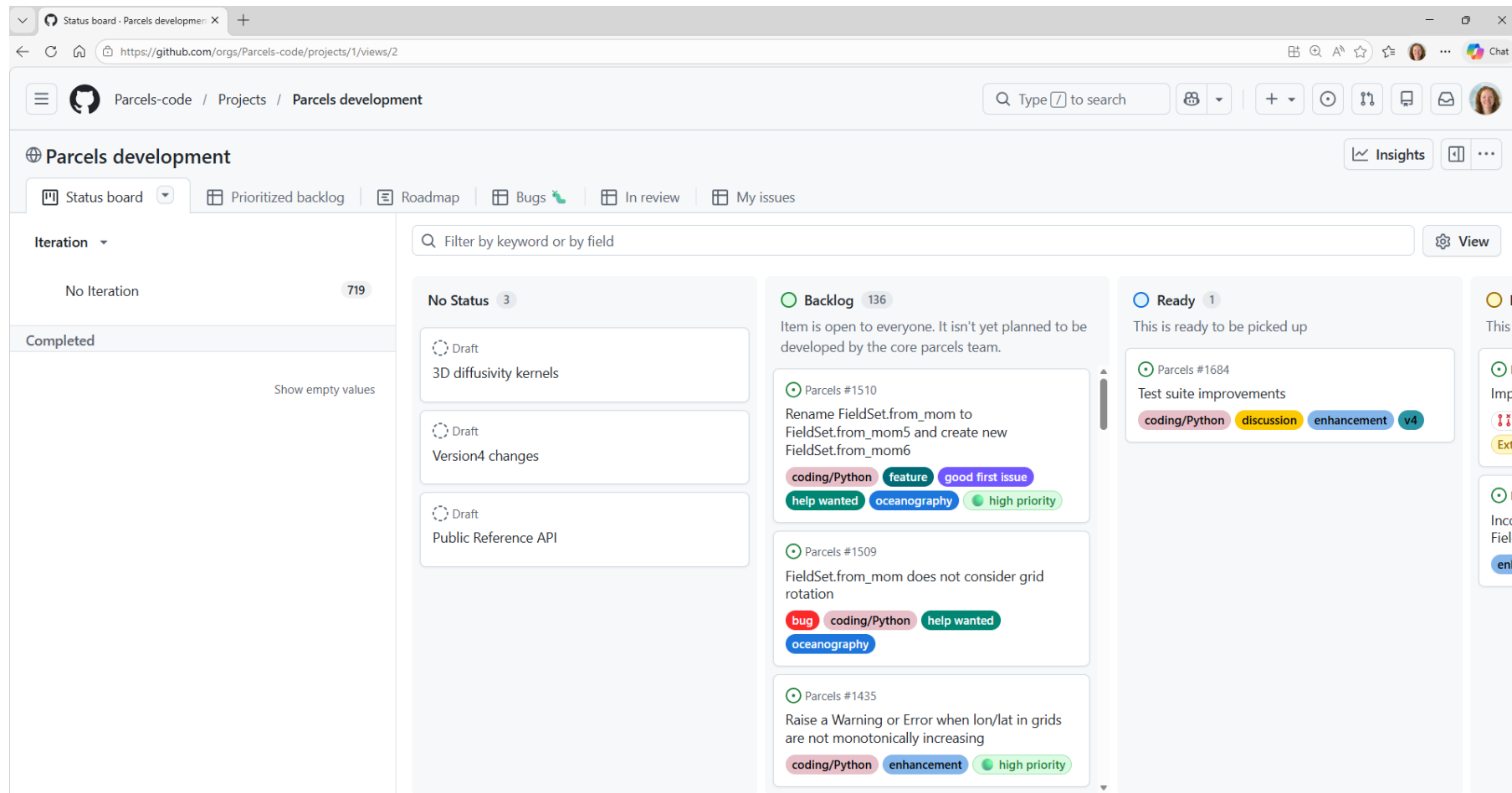


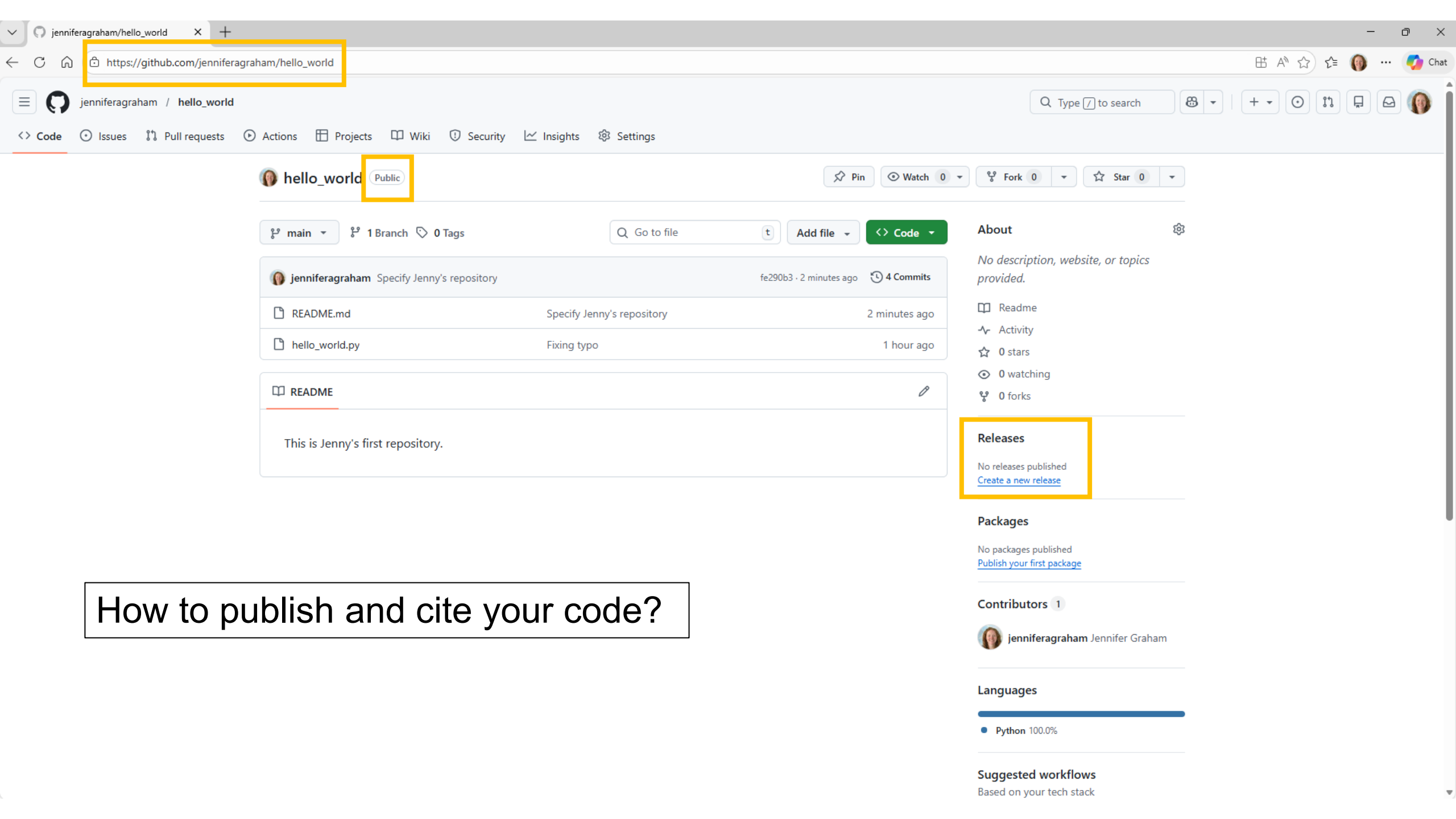
The screenshot shows the GitHub interface for the `numpy/numpy` repository. The top navigation bar includes links for Code, Issues (2.1k), Pull requests (240), Actions, Projects (3), Wiki, Security, and Insights. A search bar is present on the right. Below the navigation bar, a featured issue titled "Ecosystem compatibility with numpy 2.0" is displayed, opened by `rgommers` on April 1, 2024, with 154 comments. Below this, a filter bar shows `is:issue state:open` and buttons for Labels, Milestones, and a "New issue" button. A table of open issues follows, with columns for Open (2,067), Closed (11,230), Author, Labels, Projects, Milestones, Assignees, Types, and Newest. The table lists several issues, each with a title, a label (e.g., "03 - Maintenance", "00 - Bug", "01 - Enhancement", "04 - Documentation"), and a comment count.

Open	Closed	Author	Labels	Projects	Milestones	Assignees	Types	Newest
2,067	11,230							
Ecosystem compatibility with numpy 2.0 #26191 - rgommers opened on Apr 1, 2024 154								
MNT: new deprecations from CPython struct module in ctypes tests 03 - Maintenance #28926 - ngoldbaum opened yesterday 1								
BUG: locals() iteration bug fix for python 3.12 is missing from 1.x releases 00 - Bug #28922 - atom-andrew opened 2 days ago 4								
numpy ndindex is slower than itertools alternative 01 - Enhancement #28921 - ricardoV94 opened 2 days ago 1								
BUG: array_equal(x, y, equal_nan=True) raises for structured arrays 00 - Bug #28920 - LasseBlaauwbroek opened 2 days ago 7								
DOC: indices on numpy.fft.iff 04 - Documentation #28914 - saulopo opened 3 days ago 3								
BUG: nearest interpolation method for quantile does not match documented behaviour 04 - Documentation #28910 - wence- opened 3 days ago 4								
Avoid copy in matrix-vector dots with negative strides? 01 - Enhancement #28909 - ricardoV94 opened 3 days ago								

Projects

- Can be useful to outline planned development schedules for larger repositories, e.g., assigning work to different collaborators.





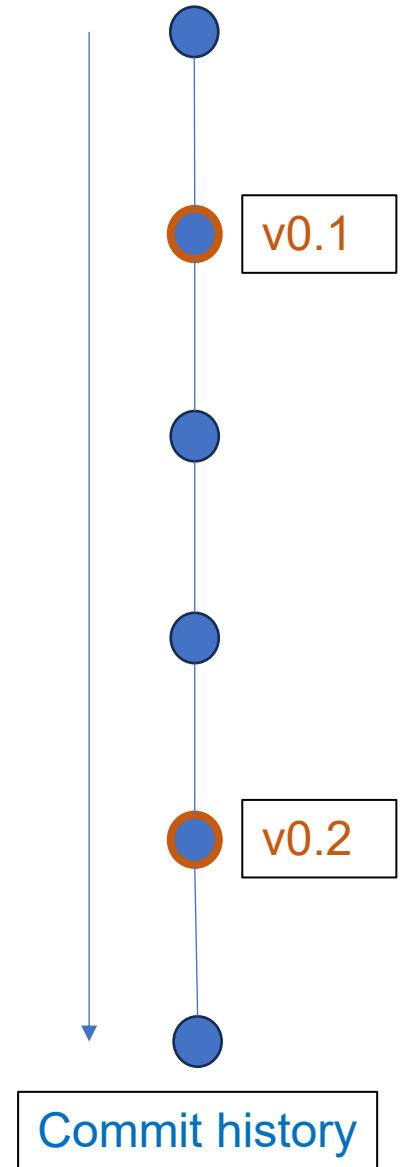
How to publish and cite your code?

GitHub Releases

Creating a release ensures that you can refer to the code at your chosen point in time.

Each release is associated with a “tag” that refers to your chosen commit in the history of your repository.

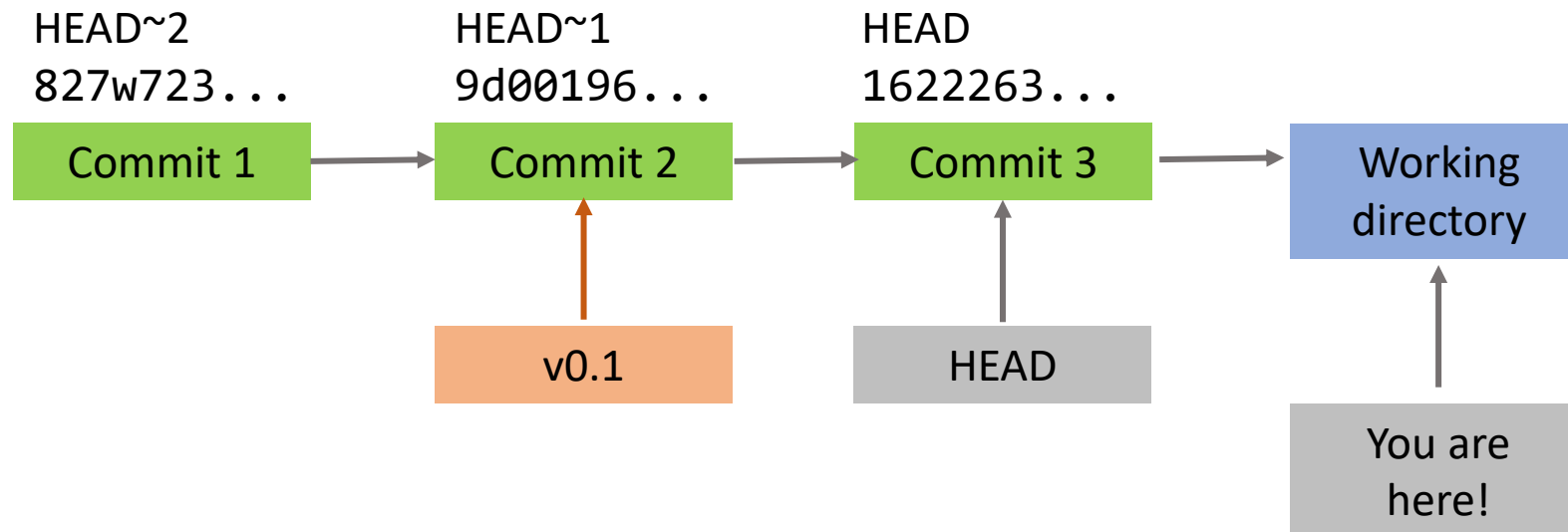
Through Zenodo, you can also automatically assign a DOI to your release on creation.

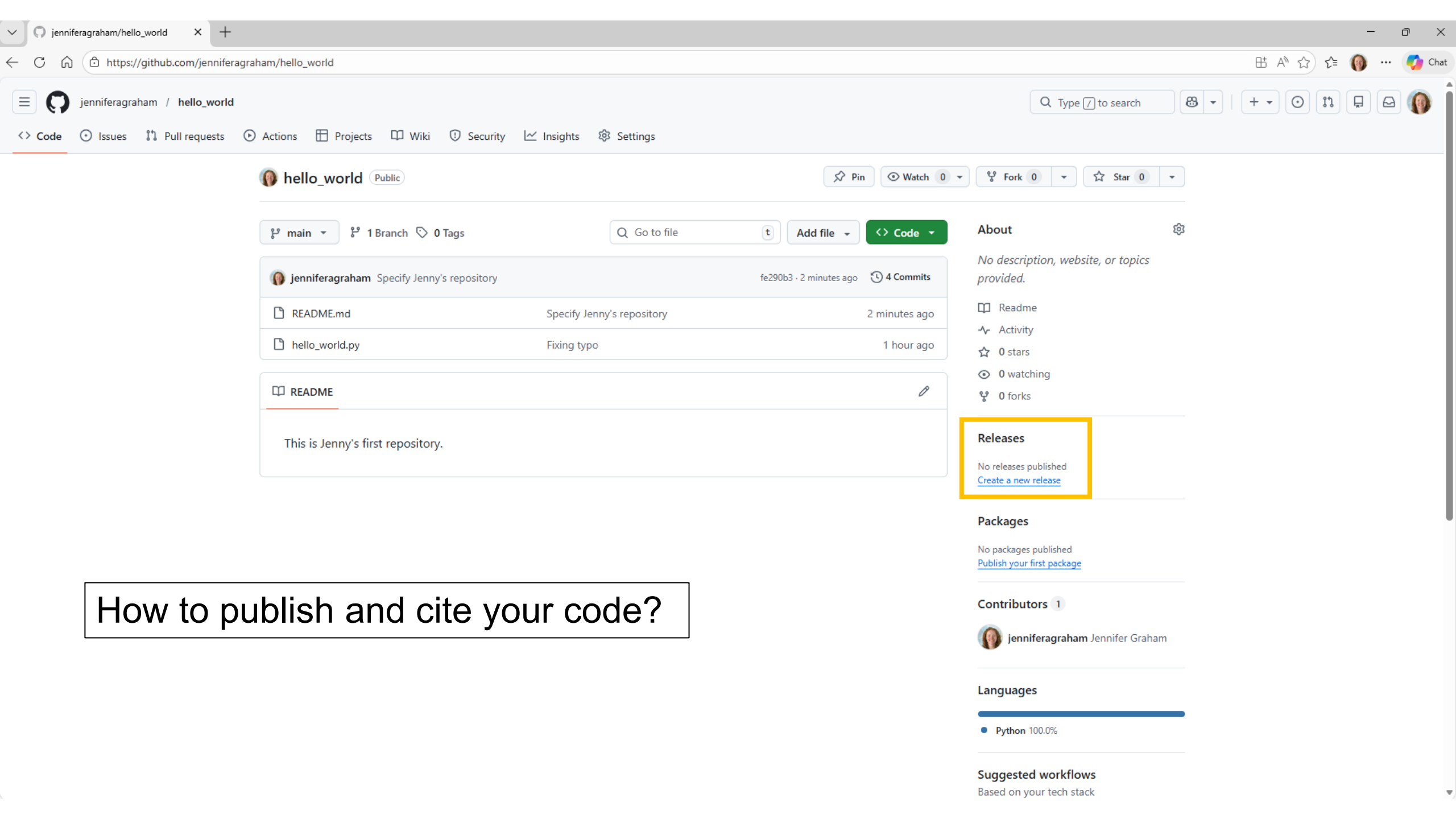


Git tag

Can also label commit versions on the command line, that you want to go back to, or release, in future.

```
git tag -a <tag-name> <commit> :: assign label <tag-name>
                                   to your commit
```





How to publish and cite your code?

New release · jenniferagraham/hello_world

https://github.com/jenniferagraham/hello_world/releases/new

Chat

jenniferagraham / hello_world

Type / to search

+ -

<> Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

← Releases

New release

Tag: v0.1

Target: main

Excellent! This tag will be created from the target when you publish this release.

Release title

First release

Release notes

Select a previous tag to create generated release notes

Previous tag: Auto

Generate release notes

Write

Preview

H B I ≡ <> 🔗 ≡ ≡ ≡ ≡ 📎 @ ↻ ←

First release for testing.

Tagging suggestions

It's common practice to prefix your version names with the letter v. Some good tag names might be v1.0.0 or v2.3.4.

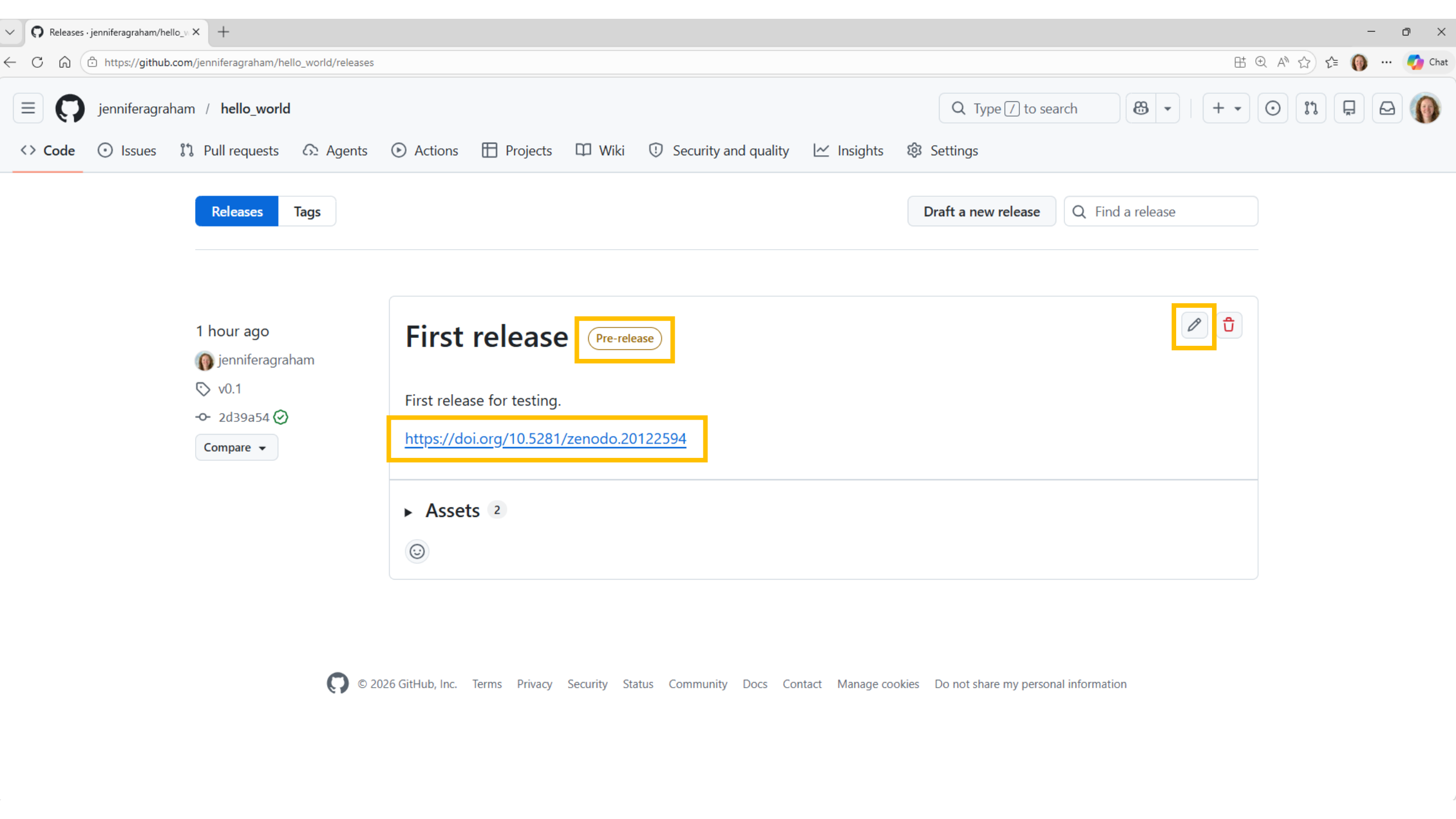
If the tag isn't meant for production use, add a pre-release version after the version name. Some good pre-release versions might be v0.2.0-alpha or v5.9-beta.3.

Semantic versioning

If you're new to releasing software, we highly recommend to [learn more about semantic versioning](#).

A newly published release will automatically be labeled as the latest release for this repository.

If 'Set as the latest release' is unchecked, the latest release will be determined by higher semantic version and creation date. [Learn more about release settings](#).



Releases Tags

Draft a new release

Find a release

1 hour ago

jenniferagraham

v0.1

2d39a54

Compare

First release

Pre-release



First release for testing.

<https://doi.org/10.5281/zenodo.20122594>

Assets 2



Creating a DOI for your release?

- Log in to Zenodo using your GitHub account, then “toggle” your repository.
 - When you sync your account, new releases will then be assigned a DOI.
 - NB. Only public repositories can be linked to Zenodo.

The image shows a two-step process for linking a GitHub repository to Zenodo. On the left, a user profile menu for 'jennifer.gr...' is open, with the 'GitHub' option highlighted. An arrow points from this menu to the main Zenodo interface on the right. The main interface is the 'My account' settings page, specifically the 'GitHub Repositories' section. It features a 'Get started' guide with three steps: 1. Flip the switch (with a toggle switch labeled 'ON'), 2. Create a release, and 3. Get the badge. Below the guide, a table lists 'Enabled Repositories' with their respective DOIs and toggle switches.

Repository	DOI	Status
CefasRepRes/CO_AMM7	DOI 10.5281/zenodo.14843574	ON
CefasRepRes/forel_uile	DOI 10.5281/zenodo.6074090	ON
jenniferagraham/hello_world	DOI 10.5281/zenodo.20122730	ON

Try it yourself [20 min]

1. Document your code!

- In your test repository (or one you made already), create a README file and/or wiki.
- Use markdown formatting, e.g., to create headings and links to files or references.
- <https://www.markdownguide.org/cheat-sheet/>

2. Create a release of your repository.

Optional:

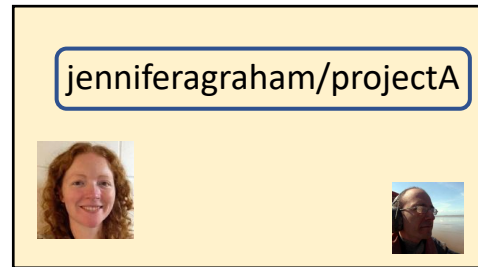
- *Create tag(s) for your chosen commit(s) on the command line.*
- *Link your account to Zenodo, to create a DOI.*

How to collaborate?

Sharing development of your code

Collaboration models

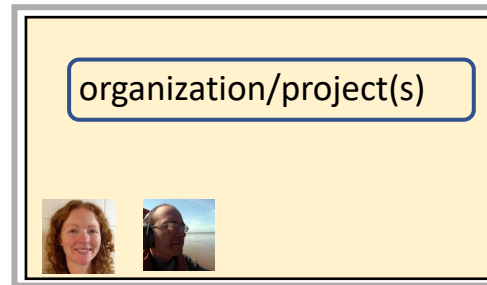
Personal account +
add collaborator



added as
collaborator



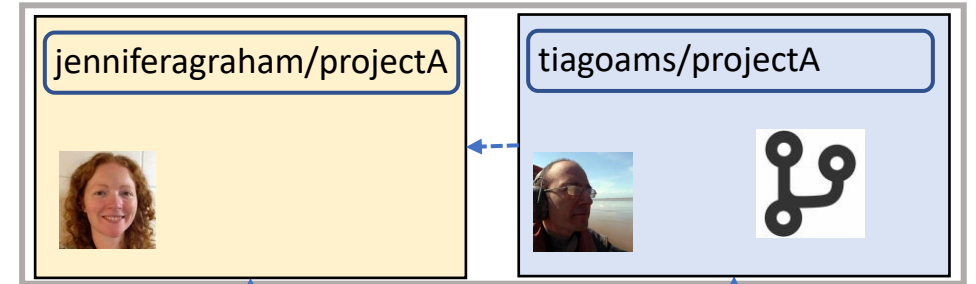
GitHub Organization



multiple
collaborators

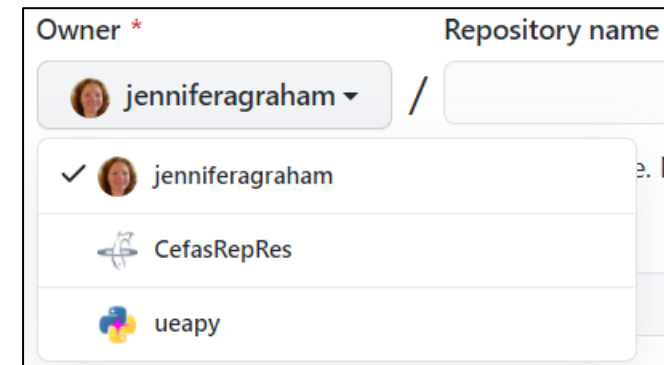


Forked + pull request
original repo forked repo



Organisations

- Organisations can add further functionality on GitHub.
 - An example from SAMS: [ScotMarPhys](#)
 - If you don't have one already, you can always [create your own...](#)
- Can develop private repositories that are still visible to all members.
- Easily share and organise work within the organisation.
- Repositories are technically “owned” by the organisation rather than person creating them.
 - NB. Select the correct “owner” on creation.



The screenshot shows the GitHub repository creation interface. At the top, there are two labels: "Owner *" and "Repository name". Below "Owner *" is a dropdown menu currently showing "jenniferaagraham". Below the dropdown is a list of suggestions: "jenniferaagraham" (with a checkmark), "CefasRepRes", and "ueapy". The "Repository name" field is empty and to the right of the owner dropdown.

ScotMarPhys

https://github.com/ScotMarPhys

Chat

ScotMarPhys

Type / to search

+ -

+

+

+

+

+

+

Overview

Repositories 8

Projects

Packages

People



ScotMarPhys

Follow

Popular repositories

m_moorproc_toolbox

Public

MATLAB based toolbox for real time and delayed mode processing of moored instrument data from OSNAP and CLASS oceanographic cruises.

MATLAB

2

glider_toolbox

Public

Forked from [socib/glider_toolbox](#)

MATLAB/Octave scripts to manage data collected by a glider fleet, including data download, data processing and product and figure generation, both in real time and delayed time.

MATLAB

ACExR_fjord_model

Public

MATLAB

ocean-glider-processing

Public

Jupyter Notebook

ScotMarPhys.OSNAP-Mooring-Processing.io

Public

HTML

ecoSUB-toolbox

Public

A toolbox for processing ecoSUB data files, including basic quality control, data conversion, and plotting of data.

Jupyter Notebook

1

People

This organization has no public members. You must be a member to see who's a part of this organization.

Top languages

MATLAB

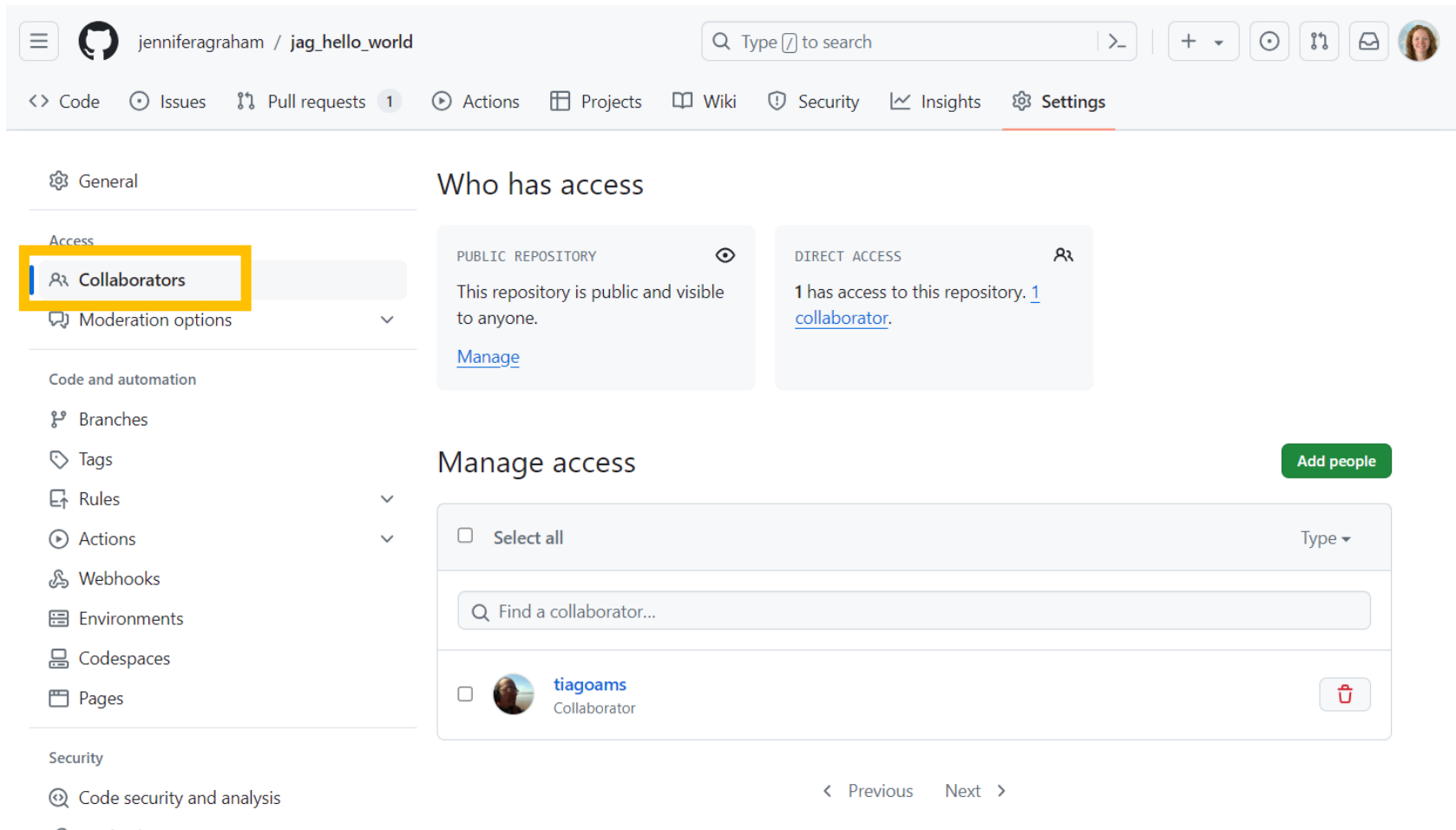
Jupyter Notebook

HTML

Report abuse

Controlling repository access...

- Collaborators can be added to each repository.



The screenshot shows the GitHub repository settings page for the repository 'jenniferagraham / jag_hello_world'. The 'Settings' tab is selected in the top navigation bar. In the left sidebar, the 'Access' section is expanded, and the 'Collaborators' option is highlighted with a yellow box. The main content area is titled 'Who has access' and displays two access types: 'PUBLIC REPOSITORY' (which is public and visible to anyone) and 'DIRECT ACCESS' (which shows that 1 collaborator has access to the repository). Below this, the 'Manage access' section is visible, featuring a search bar to find collaborators and a list of current collaborators, including 'tiagoams' who is listed as a 'Collaborator'. A green 'Add people' button is located in the top right corner of the 'Manage access' section. The bottom of the page includes navigation links for 'Previous' and 'Next'.

The screenshot shows the GitHub repository settings for 'jag_hello_world' by user 'jenniferagraham'. The 'Settings' tab is selected, and the 'Collaborators' sub-tab is active. The 'Who has access' section shows that the repository is public and has one collaborator. The 'Manage access' section lists the collaborator 'tiagoams' with a yellow highlight around the entry.

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Webhooks

Environments

Codespaces

Pages

Security

Code security and analysis

Who has access

PUBLIC REPOSITORY

This repository is public and visible to anyone.

[Manage](#)

DIRECT ACCESS

1 has access to this repository. [1 collaborator](#).

Manage access

[Add people](#)

☐ Select all Type ▾

☐	tiagoams Collaborator	
--------------------------	---------------------------------	--

[< Previous](#) [Next >](#)

- For **personal** repositories, there are two levels of permission:
 - **Repository owners** (Admin permissions)
 - **Collaborators** (aka Write access etc.)
- Only the owner can invite new collaborators.

- For **organisation** repositories, there are many levels available:

Role	Access
Admin	Full access i.e. repository owners
Maintain	Manage the repository with access to all but most sensitive settings.
Write	Can push changes to repo
Triage	Can manage issues but no write access.
Read	View code, make comments, but not change

- Only those with **Admin** access can invite new collaborators.
- Access can be grouped via “teams” as well as individuals, e.g., if regular groups require access to multiple repositories.
- NB. Outside collaborators can be invited (don’t need to be a member to collaborate)

Manage access

[Create team](#)
[Invite teams or people](#)
☐ Select all

Type ▾ Role ▾

 Find a team, organization member or outside collaborator...

☐


David Ryder
davidryderuk2

Role: Write ▾


☐


Jennifer Graham
jenniferaqram


☐


tiagoams



Choose role

Read

Can read and clone this repository. Can also open and comment on issues and pull requests.

Triage

Can read and clone this repository. Can also manage issues and pull requests.

✓ Write

Can read, clone, and push to this repository. Can also manage issues and pull requests.

Maintain

Can read, clone, and push to this repository. They can also manage issues, pull requests, and some repository settings.

Admin

Can read, clone, and push to this repository. Can also manage issues, pull requests, and repository settings, including adding collaborators.

[Security](#)
[Status](#)
[Docs](#)

[Contact GitHub](#)
[About](#)

Restricted admin access

- Within an **organisation**, repositories are technically owned by the organisation, not individual.
 - Some settings can only be changed/managed by the **organisation owner** e.g.

Danger Zone

Change repository visibility Organization members can't change repo visibility.	
Transfer ownership Organization members cannot transfer repositories	
Archive this repository Mark this repository as archived and read-only.	Archive this repository
Delete this repository Organization members cannot delete repositories.	



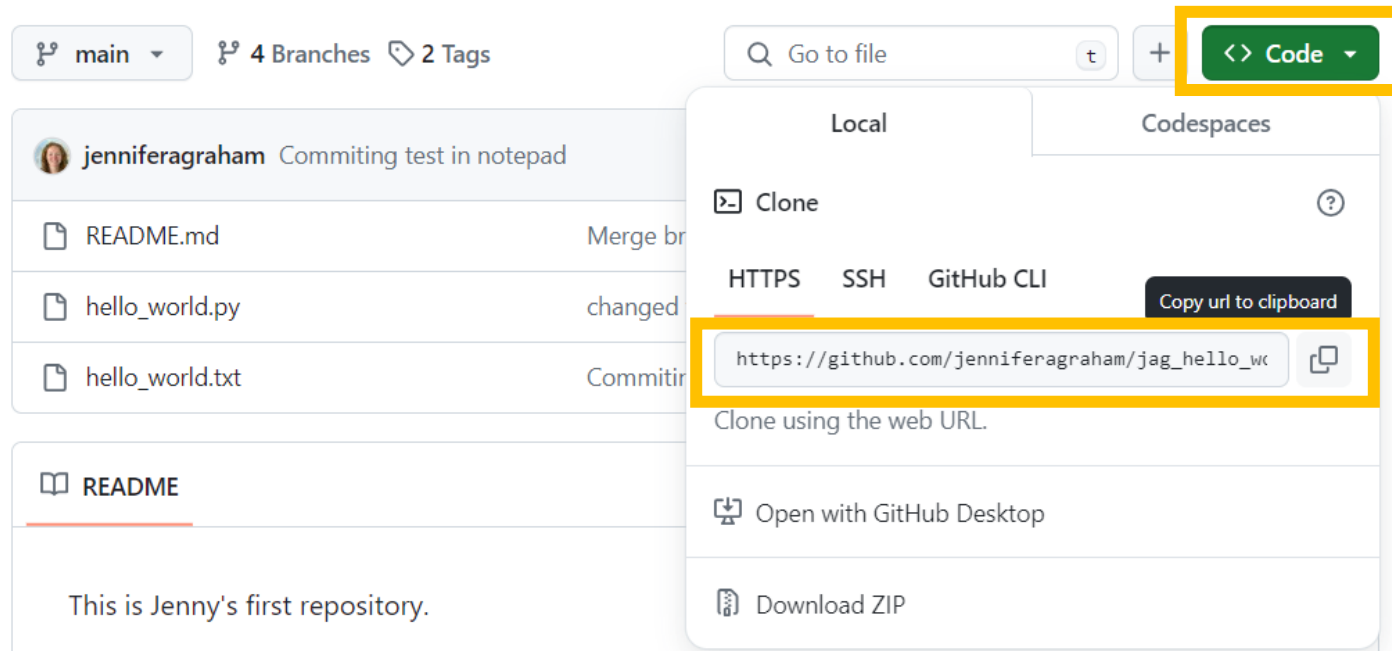
- Organisation owners can decide on the level of access for members.
- Only owners of the organisation can invite/approve new members.

(How) can I modify someone else's repository?

It depends on your access...

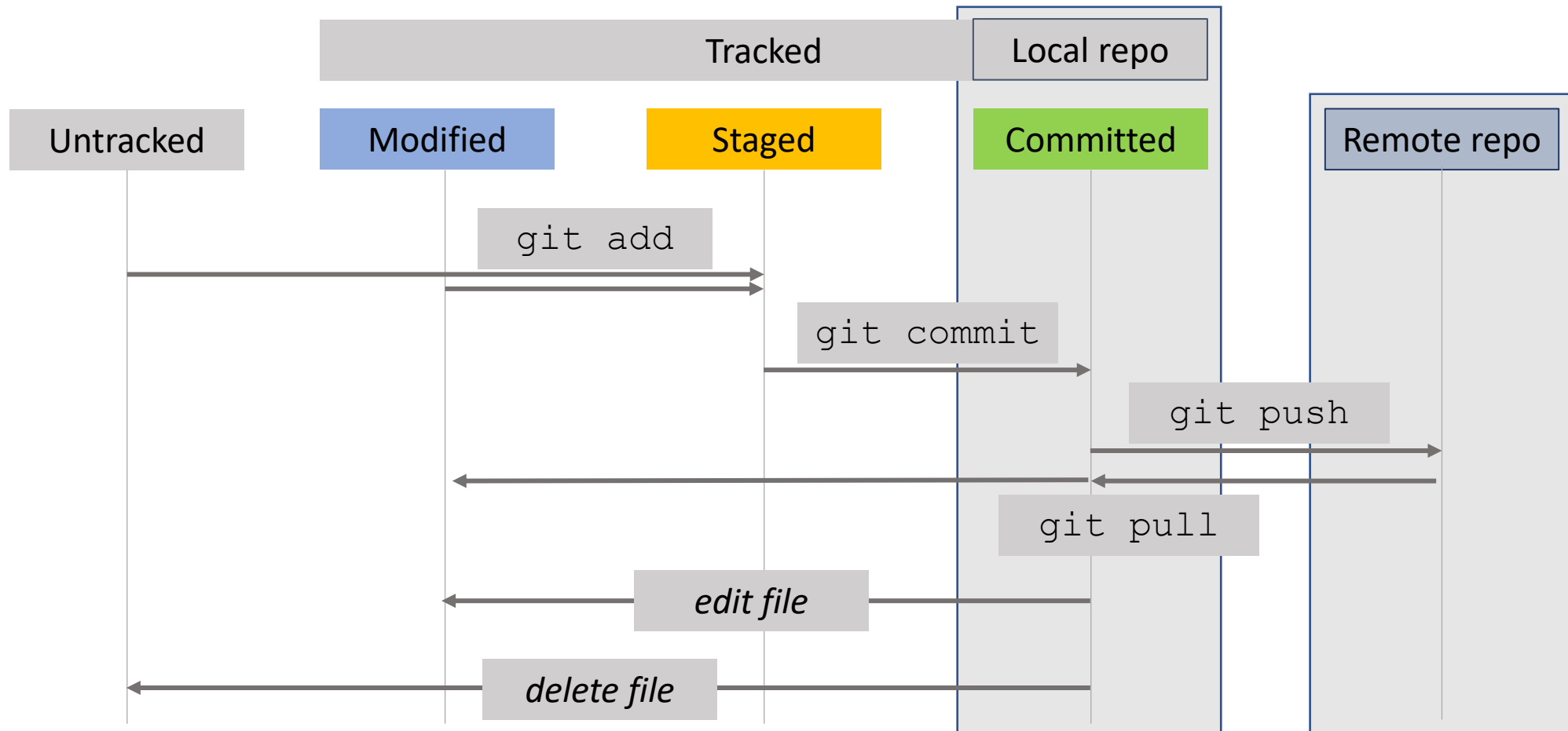
Admin, write or collaborator access...

- Use the same workflow discussed already.
 - You can push changes to main, and create branches for developments.
- Use `git clone` to make a local copy.



```
git clone <copied-url>
```

download into new local repository



If you have only read access...

e.g. For public repositories, or those shared through an organisation.

- You can still clone the repository, then track locally as normal.
- However, if you modify the repository, you won't be able to push changes back to the remote repository.
- If you want to make a copy and intend to make developments, consider forking...

What is a fork?



- Similar to a branch, in that both can be used to develop new features, away from the main branch.
- In basic terms, the difference is:
 - If you have write access for the repository -> branch.
 - If you don't have write access -> fork.
- Forking creates a copy of the original repository in your ownership (including its history).
- Repository remains linked to the original (main) so you can see how the developments diverge.
- From a fork, you must create a “pull request” to merge changes back to the original.
 - This will notify the code owner, and allow them to review changes before deciding whether or not to accept/merge (i.e. request them to pull changes into their repository).

How to create a fork?

Note: you can't create a fork owned by the same individual or organisation as the original.

The screenshot shows the GitHub interface for the repository 'jenniferagraham / jag_hello_world'. The 'Fork' button is highlighted with a yellow box, and a tooltip is visible: 'Fork your own copy of jenniferagraham/jag_hello_world to your account'. The repository details show 5 commits, 1 branch, and 0 releases. The commit history lists two commits: 'Testing commit with vi' (2 hours ago) and 'Fixing typo' (7 days ago). The README content is visible at the bottom.

← → ↻ https://github.com/jenniferagraham/jag_hello_world ☆ J ⋮

Search or jump to... / Pull requests Issues Marketplace Explore

jenniferagraham / jag_hello_world Private

Watch 0 Star 0 Fork 0

Fork your own copy of jenniferagraham/jag_hello_world to your account

Code Issues 0 Pull requests 0 Projects 0 Security Insights Settings

My first repository Edit

Manage topics

5 commits 1 branch 0 releases

Branch: master New pull request Create new file Upload files Find File Clone or download

jenniferagraham Testing commit with vi Latest commit 024d609 2 hours ago

README.md	Testing commit with vi	2 hours ago
hello_world.py	Fixing typo	7 days ago

README.md

This is Jenny's first repository test.

 CefasRepRes / jag_hello_world Private
forked from jenniferagraham/jag_hello_world

 Watch 0  Star 0  Fork 1

 Code  Pull requests 0  Projects 0  Security  Insights  Settings

My first repository Edit

[Manage topics](#)

 5 commits  1 branch  0 releases

Branch: master ▾ New pull request Create new file Upload files Find File Clone or download ▾

This branch is even with jenniferagraham:master.  Pull request  Compare

 jenniferagraham testing commit with vi Latest commit 024d609 19 hours ago		
 README.md	Testing commit with vi	19 hours ago
 hello_world.py	Fixing typo	8 days ago

 README.md 

This is Jenny's first repository test.

🔑 CefasRepRes / jag_hello_world Private
forked from jenniferagraham/jag_hello_world

👁 Watch 0 ★ Star 0 🍴 Fork 1

🔗 Code 📄 Pull requests 0 📁 Projects 0 🛡 Security 📊 Insights ⚙ Settings

My first repository

Edit

[Manage topics](#)

📦 5 commits

🌿 1 branch

📦 0 releases

Branch: master ▾

New pull request

Create new file

Upload files

Find File

Clone or download ▾

This branch is 2 commits behind jenniferagraham:master.

🔗 Pull request 🔄 Compare

 jenniferagraham testing commit with vi

Latest commit 024d609 19 hours ago

📄 README.md

Testing commit with vi

19 hours ago

📄 hello_world.py

Fixing typo

8 days ago

📖 README.md

✎

This is Jenny's first repository test.



- Pulse
- Contributors
- Traffic
- Commits
- Code frequency
- [Dependency graph](#)
- Network
- Forks**

 [jenniferagraham / jag_hello_world](#)
↳  [CefasRepRes / jag_hello_world](#)

Note: You can't create a private fork, it will always be visible to the original owner.

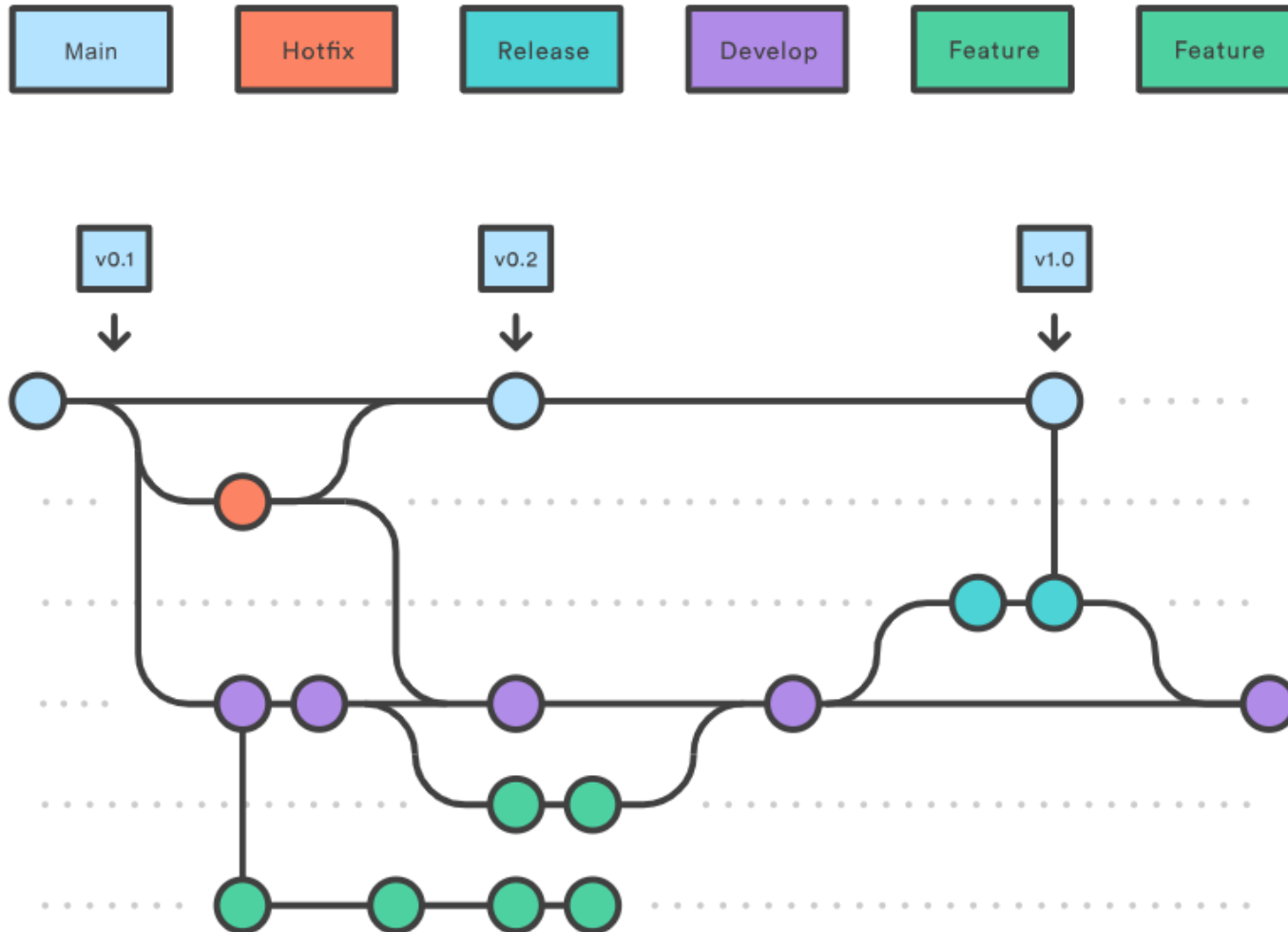
Branch	Fork
● Created in your repository ● You have total control over what you merge ● Pull requests are sent to your own repository ● Always intended to merge successful branches into master	● Creates an independent copy of another person's repository on your account so you can make changes ● You can suggest changes to the original project but they don't have to merge them ● Pull requests automatically go back to the original project ● Allows you to take a GitHub project in a different direction - may never intend to merge with original project

Collaboration workflows...

Collaboration workflows...

- git is just the tool
- Many workflows can be used, making use of various features available, and will depend on:
 - number of collaborators
 - frequency code is written
 - type of organisation (community, corporation)
 - criticality of the code

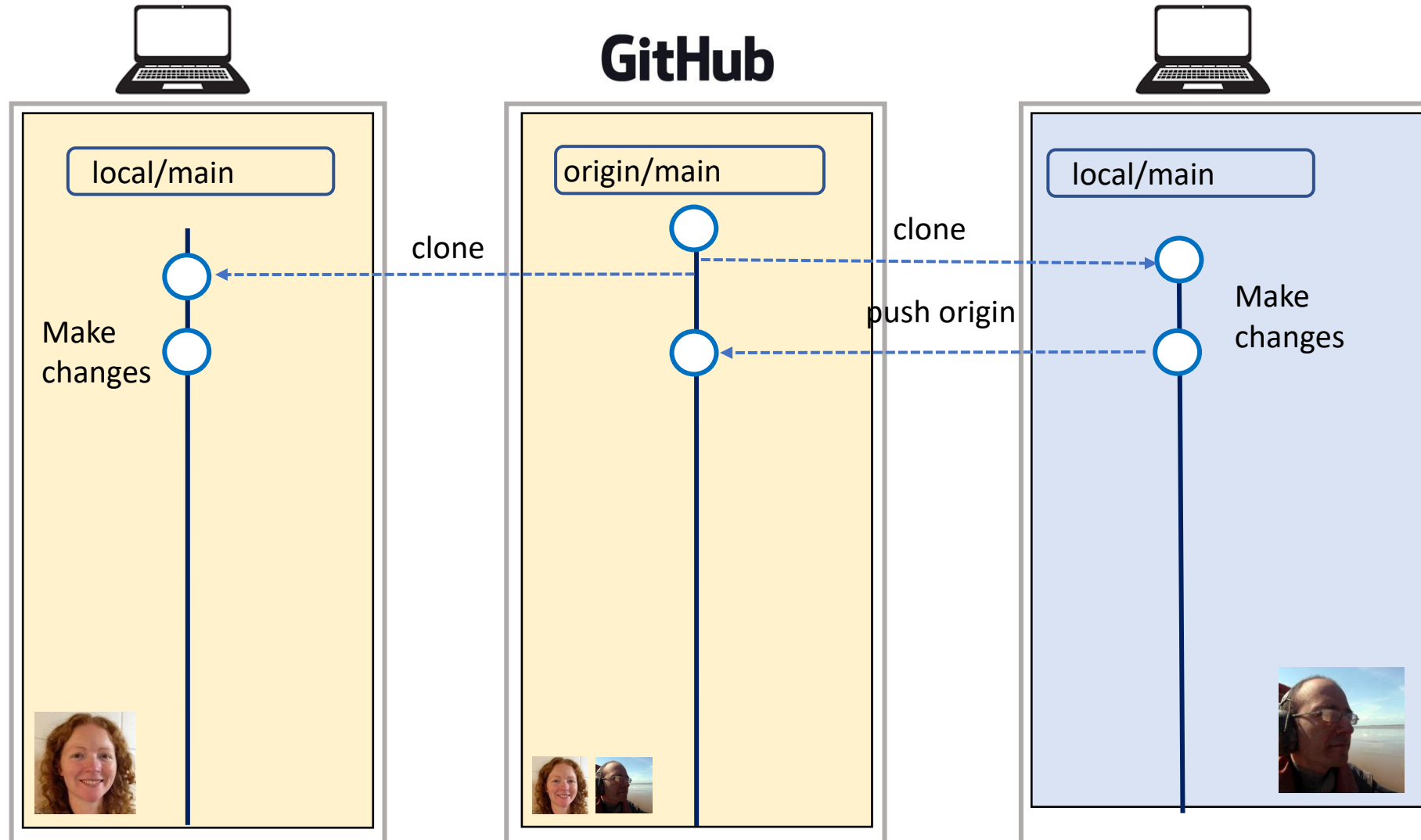
Shared access workflow, e.g., Gitflow



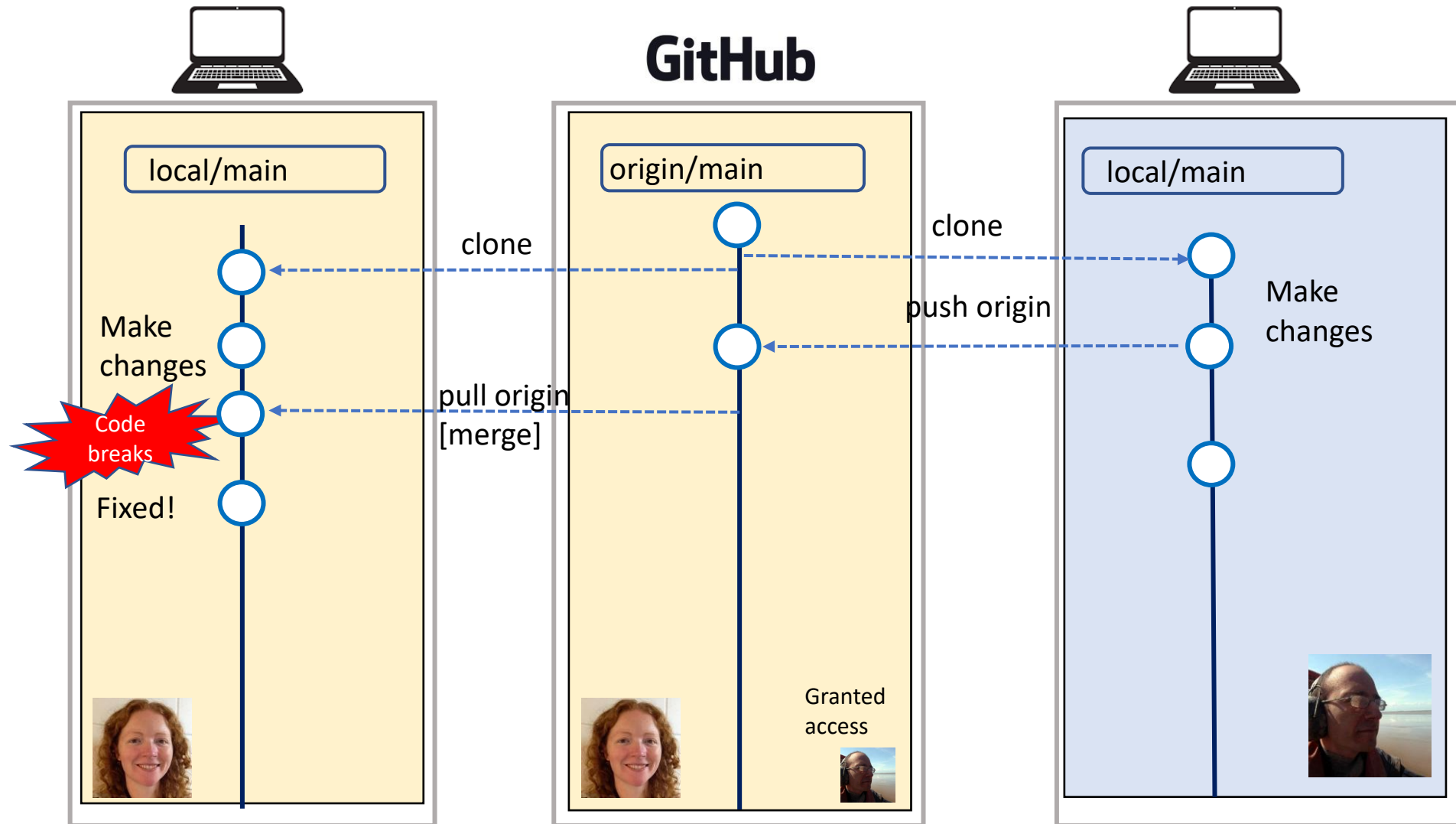
Many software developers use strict branching systems to ensure “the main is always shippable”.

Features always tested on development branch before release, so they don’t break the code!

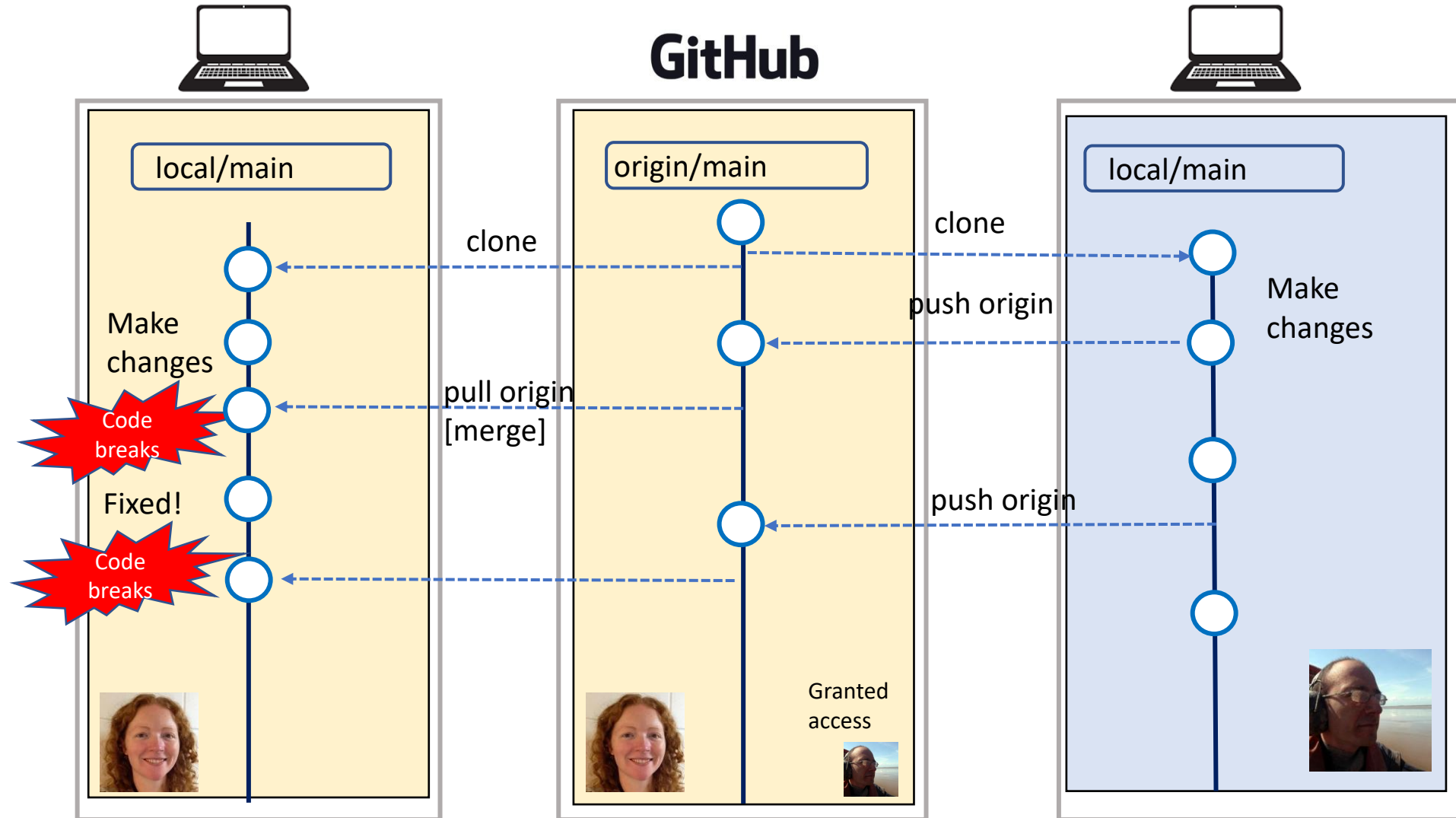
Bad workflow: competitors vs. collaborators



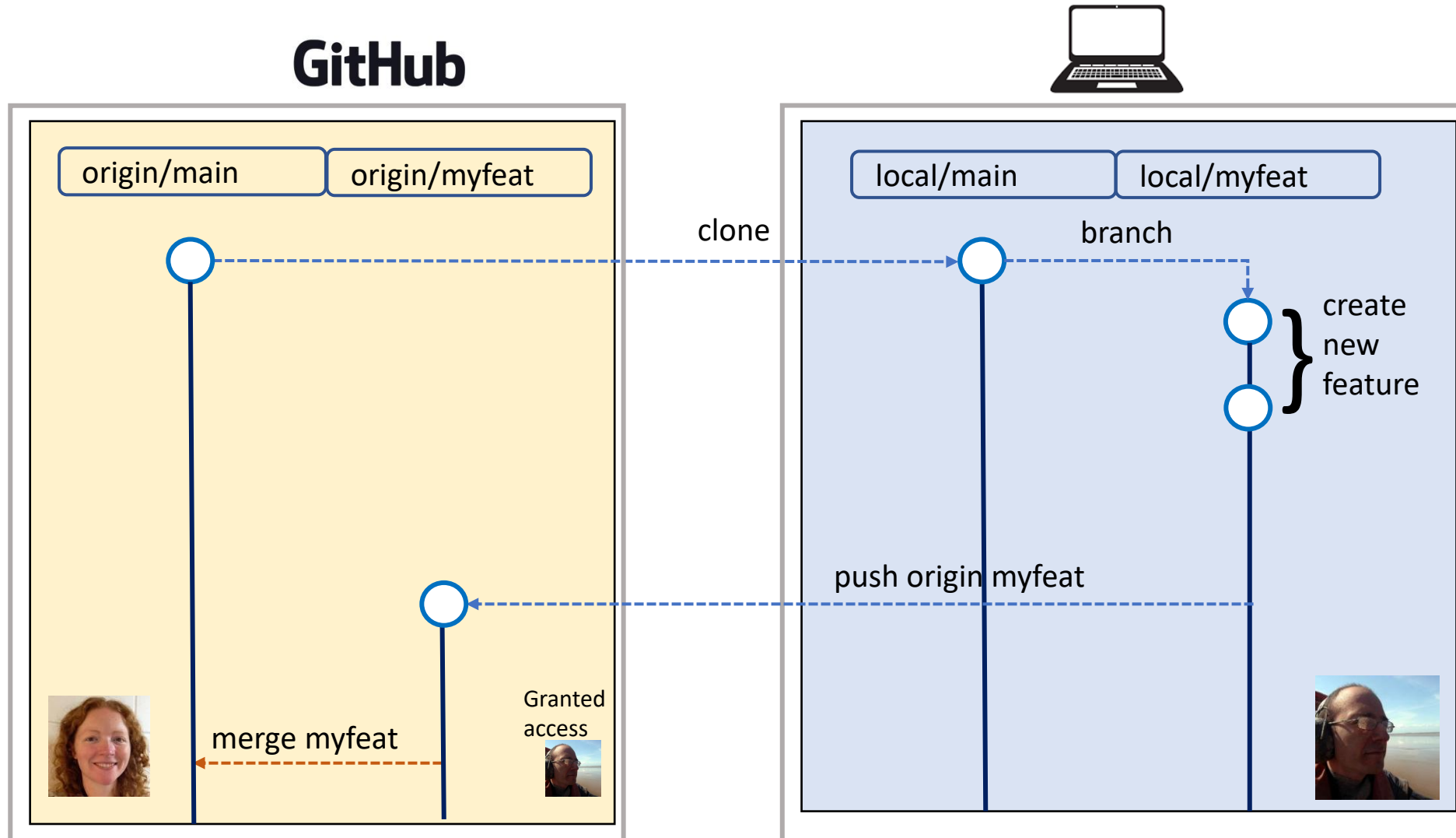
Bad workflow: competitors vs. collaborators



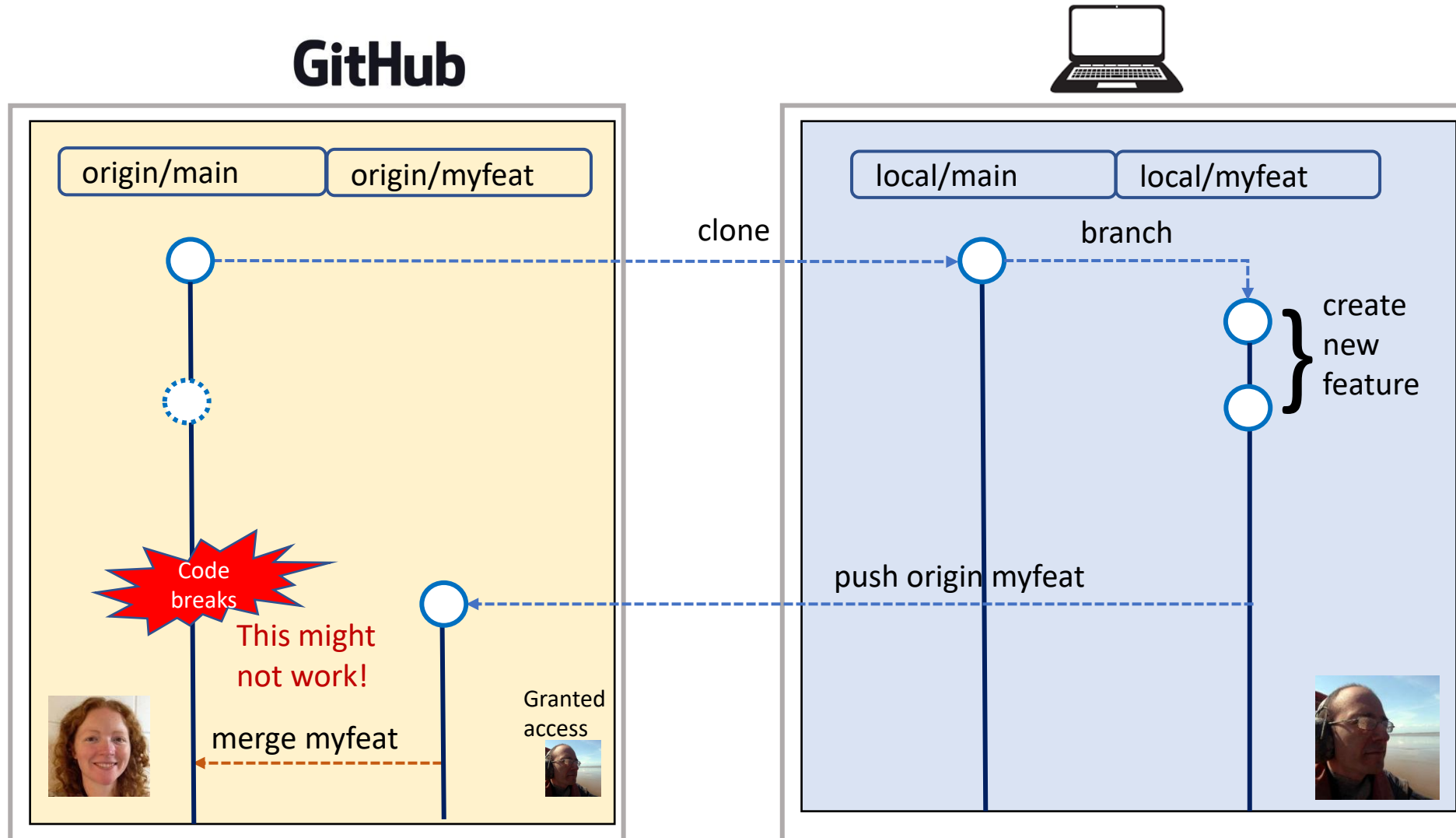
Bad workflow: competitors vs. collaborators



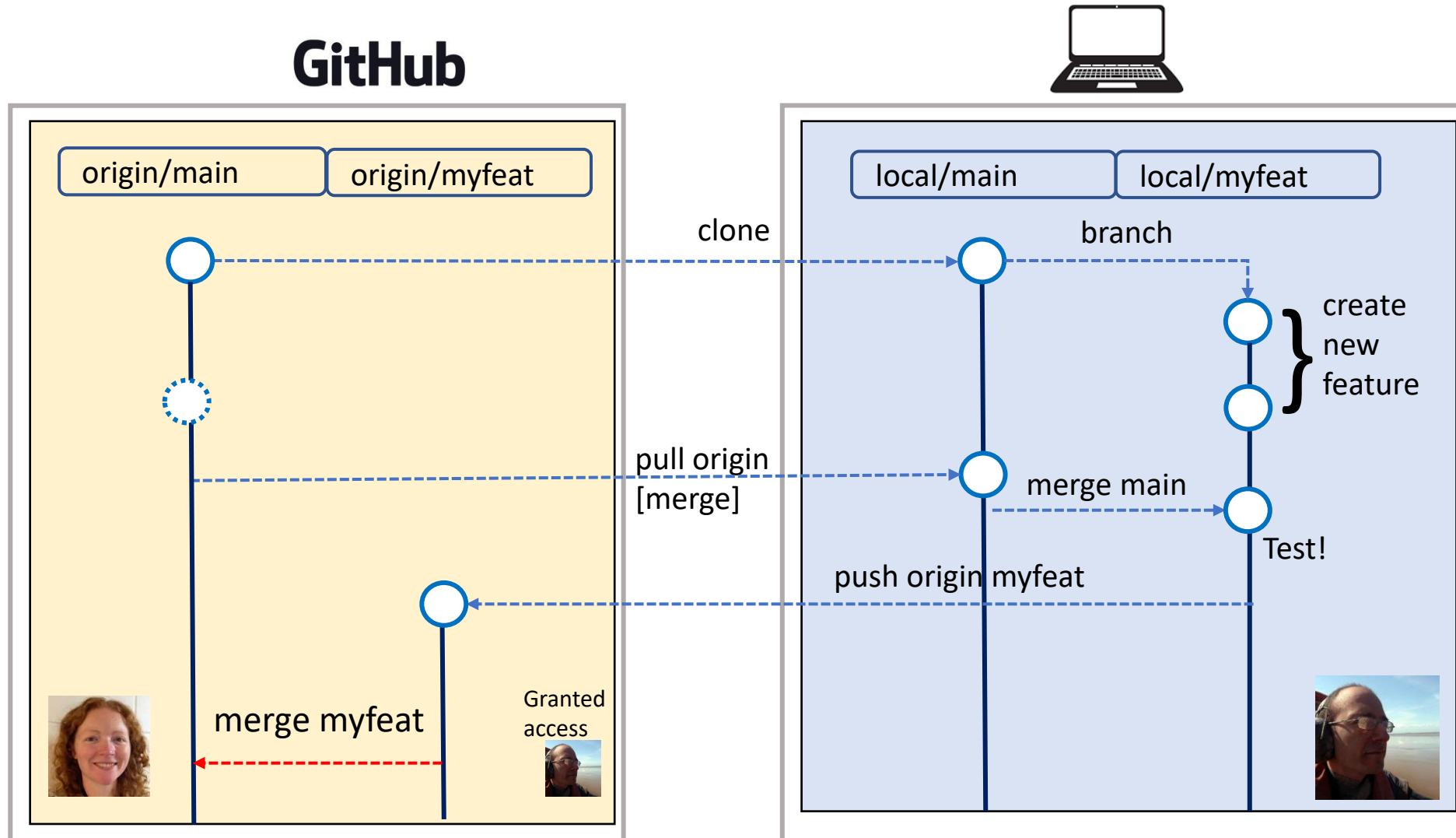
Shared workflow: Feature branches



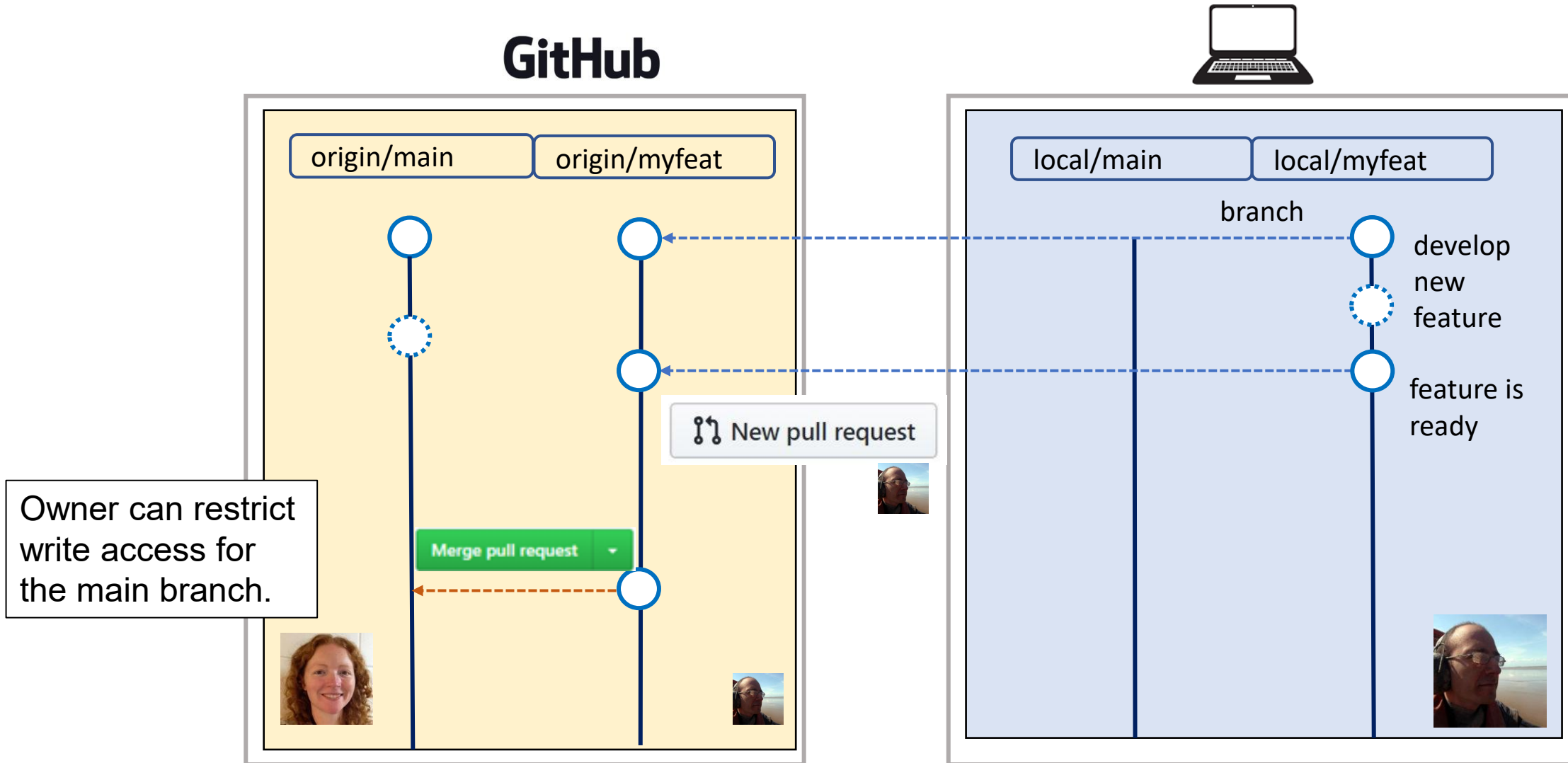
What if main branch has changed?



What if main branch has changed?



Alternative option: create a pull request



Try it yourselves [20 min]

Share your repositories!

- Pair up and find each other's test repositories.

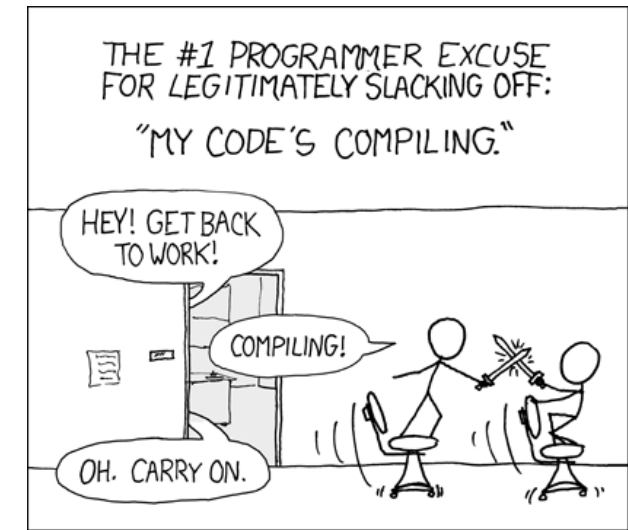
Choose how you want to collaborate?

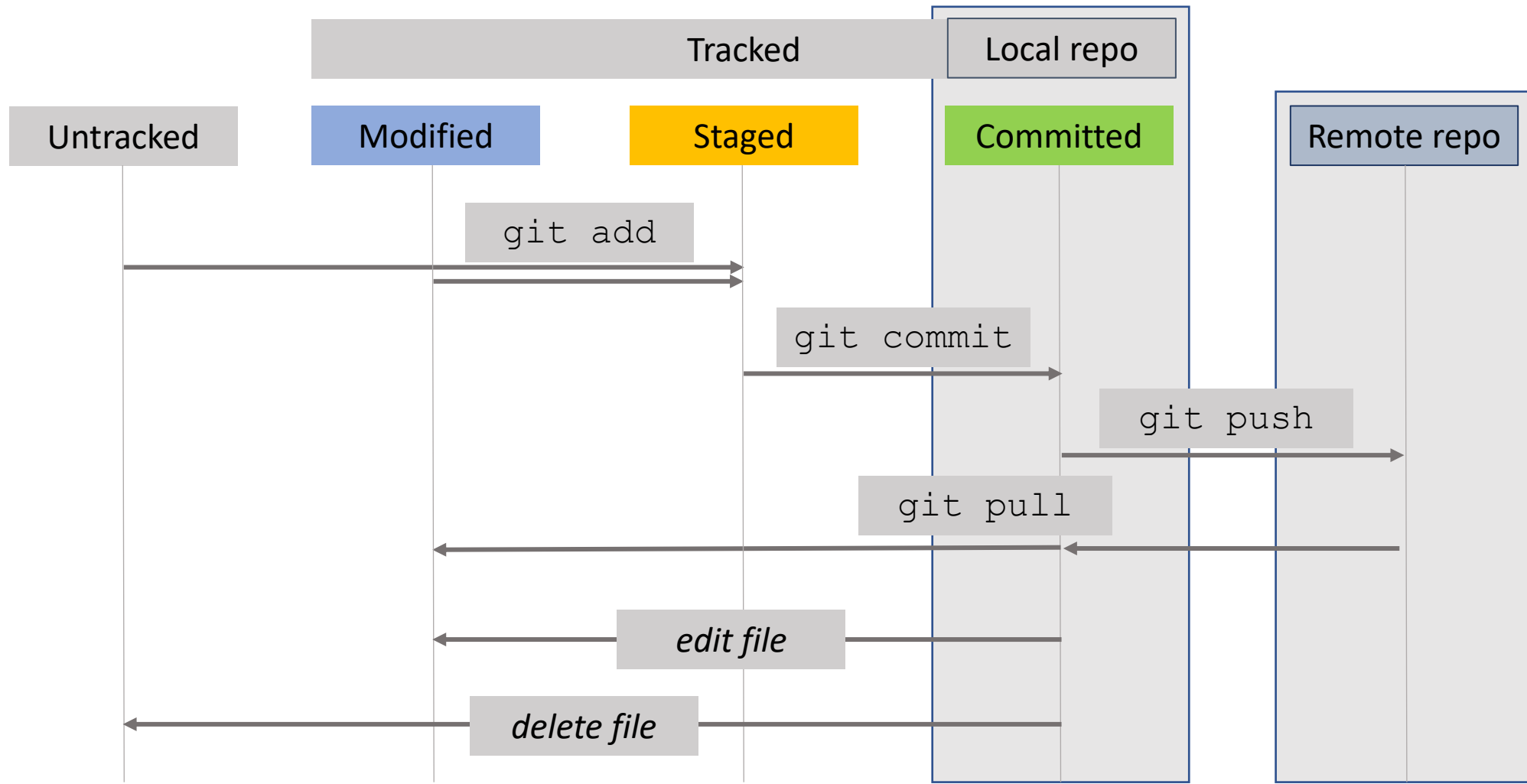
1. Provide **write access** to colleague on test repository
 - Clone, (create branches?) make commits and push changes.

Or

2. **Create a fork** and suggest changes using pull requests.
 - Clone, (create branches?) make commits and push changes
 - Create the pull request on GitHub.
 - Let the owner review changes.

When you start collaborating, check if there are any conflicts to be dealt with...?





Please provide feedback:

All comments will be anonymous

Introduction to Git & GitHub

